

Pertama kita masuk ip dan password kita dalam CMD `ssh -p 9765 Girard@103.176.78.186`. Setelah itu kita masukan code yang sudah ada pada ppt yang ada di dalam google classroom di mana yang akan pertama kita masukan yaitu ***sudo apt-get update*** code ini digunakan dalam sistem seperti ubuntu ini dapat memperbaharui informasi pada sistem. Jika code nya sudah berhasil maka kita bisa melanjutkan step selanjutnya.

```
Last login: Wed Nov 29 11:13:46 2023 from 36.85.219.228
Girard@Rampen:~$ sudo apt-get update
Hit:1 https://mirrors.idcloudhost.com/ubuntu jammy InRelease
Get:2 https://mirrors.idcloudhost.com/ubuntu jammy-updates InRelease [119 kB]
Get:3 https://mirrors.idcloudhost.com/ubuntu jammy-backports InRelease [109 kB]
Get:4 https://mirrors.idcloudhost.com/ubuntu jammy-updates/main amd64 Packages [1158 kB]
Get:5 https://mirrors.idcloudhost.com/ubuntu jammy-updates/main Translation-en [246 kB]
Get:6 https://mirrors.idcloudhost.com/ubuntu jammy-updates/main amd64 c-n-f Metadata [16.1 kB]
Get:7 https://mirrors.idcloudhost.com/ubuntu jammy-updates/restricted amd64 Packages [1106 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security InRelease [110 kB]
Get:9 https://mirrors.idcloudhost.com/ubuntu jammy-updates/restricted Translation-en [179 kB]
Get:10 https://mirrors.idcloudhost.com/ubuntu jammy-updates/restricted amd64 c-n-f Metadata [520 B]
Get:11 https://mirrors.idcloudhost.com/ubuntu jammy-updates/universe amd64 Packages [995 kB]
Get:12 https://mirrors.idcloudhost.com/ubuntu jammy-updates/universe Translation-en [218 kB]
Get:13 https://mirrors.idcloudhost.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [22.1 kB]
Get:14 https://mirrors.idcloudhost.com/ubuntu jammy-updates/multiverse amd64 Packages [41.6 kB]
Get:15 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [991 kB]
Get:16 https://mirrors.idcloudhost.com/ubuntu jammy-updates/multiverse Translation-en [9768 B]
Get:17 https://mirrors.idcloudhost.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [472 B]
Get:18 https://mirrors.idcloudhost.com/ubuntu jammy-backports/main amd64 Packages [64.2 kB]
Get:19 https://mirrors.idcloudhost.com/ubuntu jammy-backports/main Translation-en [10.5 kB]
Get:20 https://mirrors.idcloudhost.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [388 B]
Get:21 https://mirrors.idcloudhost.com/ubuntu jammy-backports/universe amd64 Packages [27.8 kB]
Get:22 https://mirrors.idcloudhost.com/ubuntu jammy-backports/universe Translation-en [16.4 kB]
Get:23 https://mirrors.idcloudhost.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [644 B]
Get:24 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [192 kB]
Get:25 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [1147 kB]
Get:26 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [187 kB]
Get:27 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [799 kB]
Get:28 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [148 kB]
Fetched 7915 kB in 4s (2256 kB/s)
Reading package lists... Done
Girard@Rampen:~$
```

Selanjutnya kita memasukan ***code sudo snap install --classic certbot*** code ini menjelaskan digunakan untuk menginstal Certbot menggunakan sistem manajemen paket Snap pada distribusi Linux yang mendukung Snap. Certbot adalah alat yang digunakan untuk mengotomatisasi proses pendaftaran dan pembaruan sertifikat SSL/TLS dari Let's Encrypt. Jika sudah terinstall kita bisa melanjutkan step berikutnya.

```
Get:14 https://mirrors.idcloudhost.com/ubuntu jammy-updates/multiverse amd64 Packages [41.6 kB]
Get:15 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [991 kB]
Get:16 https://mirrors.idcloudhost.com/ubuntu jammy-updates/multiverse Translation-en [9768 B]
Get:17 https://mirrors.idcloudhost.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [472 B]
Get:18 https://mirrors.idcloudhost.com/ubuntu jammy-backports/main amd64 Packages [64.2 kB]
Get:19 https://mirrors.idcloudhost.com/ubuntu jammy-backports/main Translation-en [10.5 kB]
Get:20 https://mirrors.idcloudhost.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [388 B]
Get:21 https://mirrors.idcloudhost.com/ubuntu jammy-backports/universe amd64 Packages [27.8 kB]
Get:22 https://mirrors.idcloudhost.com/ubuntu jammy-backports/universe Translation-en [16.4 kB]
Get:23 https://mirrors.idcloudhost.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [644 B]
Get:24 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [192 kB]
Get:25 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [1147 kB]
Get:26 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [187 kB]
Get:27 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [799 kB]
Get:28 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [148 kB]
Fetched 7915 kB in 4s (2256 kB/s)
Reading package lists... Done
Girard@Rampen:~$ sudo snap install --classic certbot
certbot 2.7.4 from Certbot Project (certbot-eff✓) installed
Girard@Rampen:~$
```

Selanjutnya kita masukan **code `sudo certbot --help`** digunakan untuk mendapatkan bantuan (help) terkait dengan penggunaan Certbot. Dengan menjalankan perintah ini, kita bisa mendapatkan informasi tentang opsi dan argumen yang dapat digunakan bersama dengan Certbot. Jika sudah berhasil kita bisa melanjutkan step berikutnya.

```
Girard@Rampen:~$ sudo certbot --help
-----
certbot [SUBCOMMAND] [options] [-d DOMAIN] [-d DOMAIN] ...

Certbot can obtain and install HTTPS/TLS/SSL certificates.  By default,
it will attempt to use a webserver both for obtaining and installing the
certificate.  The most common SUBCOMMANDS and flags are:

obtain, install, and renew certificates:
    (default) run          Obtain & install a certificate in your current webserver
    certonly              Obtain or renew a certificate, but do not install it
    renew                 Renew all previously obtained certificates that are near
expiry
    enhance               Add security enhancements to your existing configuration
    -d DOMAINS            Comma-separated list of domains to obtain a certificate for

    --apache              Use the Apache plugin for authentication & installation
    --standalone           Run a standalone webserver for authentication
    --nginx               Use the Nginx plugin for authentication & installation
    --webroot             Place files in a server's webroot folder for authentication
    --manual              Obtain certificates interactively, or using shell script
hooks
    -n                   Run non-interactively
    --test-cert           Obtain a test certificate from a staging server
    --dry-run             Test "renew" or "certonly" without saving any certificates
to disk

manage certificates:
    certificates          Display information about certificates you have from Certbot
    revoke               Revoke a certificate (supply --cert-name or --cert-path)
    delete              Delete a certificate (supply --cert-name)
    reconfigure          Update a certificate's configuration (supply --cert-name)

manage your account:
    register             Create an ACME account
    unregister           Deactivate an ACME account
    update_account        Update an ACME account
Girard@Rampen:~$
    --agree-tos          Agree to the ACME server's Subscriber Agreement
    -m EMAIL             Email address for important account notifications

More detailed help:
```

Selanjutnya kita masukan **code `sudo certbot --apache`** digunakan bersama dengan Certbot untuk mengotomatisasi proses pendaftaran dan pembaruan sertifikat SSL/TLS dari Let's Encrypt, khususnya ketika Anda menggunakan server web Apache. Di sini kita bisa masukan gmail kita jadi saya masukan gmail student saya s2200288@student.unklab.ac.id. Jika sudah memasukan gmail kita dan domain name kita, selanjutnya kita klik huruf Y jika sudah kita bisa melanjutkan program kita.

```

-h, --help [TOPIC]    print this message, or detailed help on a topic;
                      the available TOPICS are:

    all, automation, commands, paths, security, testing, or any of the
    subcommands or plugins (certonly, renew, install, register, nginx,
    apache, standalone, webroot, etc.)
-h all                print a detailed help page including all topics
--version             print the version number
-----
irard@Rampen:~$ sudo certbot --apache
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Enter email address (used for urgent renewal and security notices)
(Enter 'c' to cancel): s2200288@student.unklab.ac.id

-----
Please read the Terms of Service at
https://letsencrypt.org/documents/LE-SA-v1.3-September-21-2022.pdf. You must
agree in order to register with the ACME server. Do you agree?
-----
Y)es/(N)o: y

-----
Would you be willing, once your first certificate is successfully issued, to
share your email address with the Electronic Frontier Foundation, a founding
partner of the Let's Encrypt project and the non-profit organization that
develops Certbot? We'd like to send you email about our work encrypting the web,
EFF news, campaigns, and ways to support digital freedom.
-----
Y)es/(N)o: y
Account registered.
Please enter the domain name(s) you would like on your certificate (comma and/or
space separated) (Enter 'c' to cancel):

```

Masi code yang sama ***sudo certbot --apache*** sama seperti yang di atas disini juga kita harus memasukan domain name kita yang kita sudah beli di ***niagahoster*** yaitu ***unkgir.online***

```
Girard@Rampen:~$ sudo certbot --apache
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Please enter the domain name(s) you would like on your certificate (comma and/or
space separated) (Enter 'c' to cancel): unkgir.online
Requesting a certificate for unkgir.online

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/unkgir.online/fullchain.pem
Key is saved at: /etc/letsencrypt/live/unkgir.online/privkey.pem
This certificate expires on 2024-02-27.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for unkgir.online to /etc/apache2/sites-available/000-default-le-ssl.conf
Congratulations! You have successfully enabled HTTPS on https://unkgir.online

- - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt: https://letsencrypt.org/donate
 * Donating to EFF: https://eff.org/donate-le
- - - - -
Girard@Rampen:~$
```

Saat ini saya ingin mengubah background dalam web, dan saya mengantinnnya dengan manual di CMD.

```
Last login: Wed Nov 29 12:24:42 2023 from 36.85.219.228
Girard@Rampen:~$ sudo su
root@Rampen:/home/Girard# cd ..
root@Rampen:/home# cd ..
root@Rampen:/# ls
bin    dev    home   lib    lib64  lost+found  mnt  proc  run  snap  sys  usr
boot  etc    jenkins.war  lib32  libx32  media      opt  root  sbin  srv   tmp  var
root@Rampen:/# cd var
root@Rampen:/var# ls
backups  cache  crash  lib  local  lock  log  mail  opt  run  snap  spool  tmp  www
root@Rampen:/var# cd www/html/
root@Rampen:/var/www/html# ls
devops  index.html  unkpresent
root@Rampen:/var/www/html# cd unkpresent/
root@Rampen:/var/www/html/unkpresent# ls
app  public
root@Rampen:/var/www/html/unkpresent# cd public
root@Rampen:/var/www/html/unkpresent/public# cd ..
root@Rampen:/var/www/html/unkpresent# cd app
root@Rampen:/var/www/html/unkpresent/app# ls
config  controller  core  init.php  model  view
root@Rampen:/var/www/html/unkpresent/app# cd config
root@Rampen:/var/www/html/unkpresent/app/config# ls
config.php
root@Rampen:/var/www/html/unkpresent/app/config# sudo nano config.php
root@Rampen:/var/www/html/unkpresent/app/config# sudo cat
^Z
[1]+  Stopped                  sudo cat
root@Rampen:/var/www/html/unkpresent/app/config# sudo cat config.php
<?php

    // define path constants
    define("APP_PATH", "https://unkgir.online/unkpresent/public");

    // define database config
    define("HOST_DB", "localhost");
    define("NAME_DB", "db_attendance");
```

Setelah itu code tersebut kita ubah **http** menjadi **https** dan **ip server** kita ganti menjadi **unkgir.online**. setelah itu kita copy <https://unkgir.online/unkpresent/public> setelah kita copy kita masukan copianitu ke dalam crome dan kita klik enter.

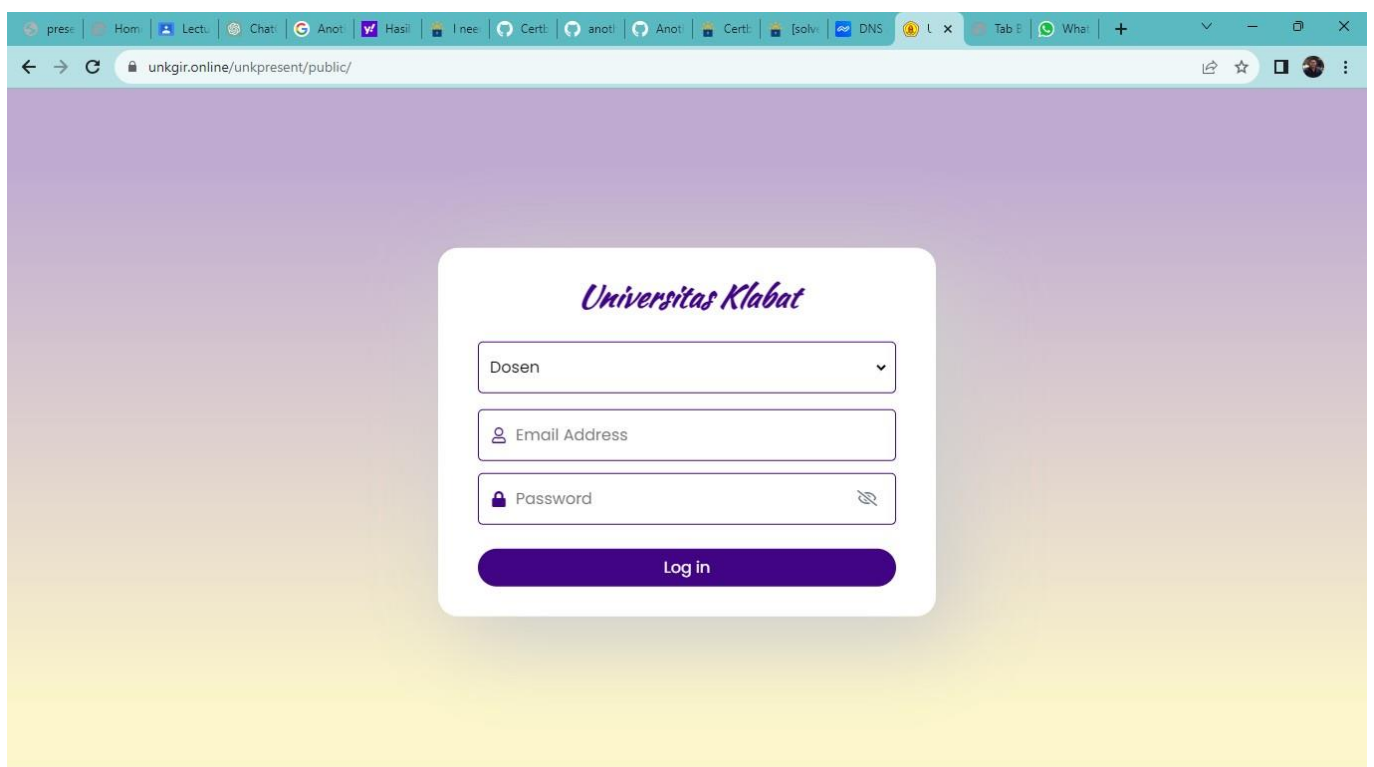
```
root@Rampen: /var/www/html/unkpresent/app/config
GNU nano 6.2 config.php *
<?php

    // define path constants
define("APP_PATH", "https://unkgir.online/unkpresent/public");

// define database config
define("HOST_DB", "localhost");
define("NAME_DB", "db_attendance");
define("USER_DB", "girard");
define("PASS_DB", "Girard18!");

?>
```

Selanjutnya ketika kita enter copian yang ada di atas kita tunggu beberapa detik, setelah kita tunggu akan ada tampilan seperti yang ada di gambar setelah itu kita coba masuk ke dalam web tersebut, kita masuk sebagai admin john.



jika kita sudah masuk sebagai admin maka tampilan yang akan keluar seperti yang ada di gambar, di mana jika tampilan seperti yang ada di gambar maka sudah berhasil. Dimana kita masuk sebagai admin John Doe email john@example.com. Sekian dari report project saya Terima Kasih.

